employing powered components for normal directional control should provide a secondary means of actuating the device unless an alternative system is provided.
The secondary means of actuating the directional control device may

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SOURCE: MSC.36(63).pdf
employing powered components for normal directional control should provide a secondary means of actuating the device unless an alternative system is provided.
The secondary means of actuating the directional control device may be manually driven when the Administration is satisfied that this is adequate, bearing in mind the craft's size and design and any limitations of speed or other parameters that may be necessary.
The directional control systems should be constructed so that a single failure in one drive or system, as appropriate, will not render any other one inoperable or unable to bring the craft to a safe situation. The Administration may allow a short period of time to permit the connection of a secondary control device when the design of the craft is such that such delay will not, in their opinion, hazard the craft.
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5.2.5 A failure mode and effect analysis should include the directional control system.
5.2.6 If necessary to bring the craft to a safe condition, power drives

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5.2.5 A failure mode and effect analysis should include the directional control system.
5.2.6 If necessary to bring the craft to a safe condition, power drives for directional control devices, including those required to direct thrust forward or astern, should become operative automatically, and respond correctly, within 5 s of power or other failure. Back-up electrical systems may be required for the starting-up time of an auxiliary diesel according to 12.2 or an emergency diesel generator according to 12.3.6.
5.2.7 Directional control devices involving variable geometry of the craft or its lift system components should, so far as is practicable, be so constructed that any failure of the drive linkage or actuating system will not significantly hazard the craft.
5.3 Demonstrations
5.3.1 The limits of safe use of any of the control system devices should

be based on demonstrations and verification process in accordance with annex 8.
5.3.2 Demonstration in accordance with annex 8 should determine any adverse effects upon safe operation of the craft in the event of an

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be based on demonstrations and verification process in accordance with annex 8.
5.3.2 Demonstration in accordance with annex 8 should determine any adverse effects upon safe operation of the craft in the event of an uncontrollable total deflection of anyone control device. Any limitation on the operation of the craft as may be necessary to ensure that the redundancy or safeguards in the systems provide equivalent safety should be included in the craft operating manual.
5.4 Control position
5.4.1 All directional control systems should normally be operated from the craft's operating station.
5.4.2
5.4.3
If directional control systems can also be operated from other positions, then two-way communication should be arranged between the operating station and these other positions.
Adequate indications should be provided at the operating station and these other positions to provide the person controlling the craft with verification of the correct response of the directional control device to this demand, and also to indicate any abnormal responses or malfunction. The indications of steering response or rudder

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with verification of the correct response of the directional control device to this demand, and also to indicate any abnormal responses or malfunction. The indications of steering response or rudder angle indicator should be independent of the system for directional control. The logic of such feedback and indications should be consistent with the other alarms and indications so that in an emergency operators are unlikely to be confused.
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CHAPTER 6 - ANCHORING, TOWING AND BERTHING
6.1 General
6.1.1 A primary assumption made in this chapter is that high speed craft will only need an anchor for emergency purposes.
6.1.2 The arrangements for anchoring, towing and berthing and the local craft structure, the design of the anchor, towing and berthing arrangements and the local craft structure should be such that risks

to persons carrying out anchoring, towing or berthing procedures are kept to a minimum.
6.1.3 All anchoring equipment, towing bitts, mooring bollards, fairleads,

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arrangements and the local craft structure should be such that risks to persons carrying out anchoring, towing or berthing procedures are kept to a minimum.
6.1.3 All anchoring equipment, towing bitts, mooring bollards, fairleads, cleats and eyebolts should be so constructed and attached to the hull that in use up to design loads, the watertight integrity of the craft will not be impaired. Design loads and any directional limitations assumed should be listed in the craft operating manual.
6.2 Anchoring
6.2.1 High speed craft should be provided with at least one anchor with its associated cable or cable and warp and means of recovery. Every craft should be provided with adequate and safe means for releasing the anchor, its cable and warp.
6.2.2 Good engineering practice should be followed in the design of any enclosed space containing the anchor recovery equipment to ensure that persons using the equipment are not put at risk. Particular care should be taken with the means of access to such spaces, the walkways, the illumination and protection from the cable and the recovery machinery.
6.2.3 Adequate arrangements should be provided for two-way voice

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care should be taken with the means of access to such spaces, the walkways, the illumination and protection from the cable and the recovery machinery.
6.2.3 Adequate arrangements should be provided for two-way voice communication between the operating compartment and persons engaged in dropping, weighing or releasing the anchor.
6.2.4 The anchoring arrangements should be such that any surfaces against which the cable may chafe (for example, hawse pipes and hull obstructions) are designed to prevent the cable from being damaged and fouled. Adequate arrangements should be provided to secure the anchor under all operational conditions.
6.2.5 The craft should be protected so as to minimize the possibility of the anchor and cable damaging the structure during normal operation.
6.3 Towing
6.3.1 Adequate arrangements should be provided to enable the craft to be towed in the worst intended conditions. Where towage is to be from more than one point a suitable bridle should be provided.
6.3.2 The towing arrangements should be such that any surfaces against which the towing cable may chafe (for example, fairleads), is of

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more than one point a suitable bridle should be provided.
6.3.2 The towing arrangements should be such that any surfaces against which the towing cable may chafe (for example, fairleads), is of sufficient radius to prevent the cable being damaged when under load.

6.3.3 The maximum permissible speed at which the craft may be towed should be included in the operating manual.
6.4 Berthing
6.4.1 Where necessary, suitable fairleads, bitts and mooring ropes should be provided.
6.4.2 Adequate storage space for mooring lines should be provided such that they are readily available and secured against the high relative wind speeds and accelerations which may be experienced.